Chemical Nomenclature

Honors Chemistry Notes: naming compounds and writing formulas

Ionic vs. covalent: A metal + nonmetal (or a polyatomic ion) forms an **ionic** compound. Two nonmetals form a **covalent (molecular)** compound. Acids are hydrogen compounds that produce H^+ in water.

1. Binary Ionic Compounds

A binary ionic compound contains exactly two elements: a metal (cation) and a nonmetal (anion). Use this section for metals with only one possible charge (Groups 1, 2, and Al, Zn, Ag).

Naming

1. Name the metal (cation) first, using its element name unchanged.
2. Name the nonmetal (anion) second, changing the ending to **-ide**.
3. Do not use prefixes. The charges already determine the ratio.

Writing the formula

4. Write the symbol and charge of each ion (cation first).
5. Balance the charges: decide how many of each ion are needed so the total positive charge equals the total negative charge. Use the smallest numbers that work.
6. Write those numbers as subscripts. Leave out subscripts of 1.
7. Check that the total positive charge equals the total negative charge.

Common charges

Group	Charge	Examples
1	1+	Li^+, Na^+, K^+
2	2+	Mg^{2+}, Ca^{2+}, Ba^{2+}
13	3+	Al^{3+}
15	3−	N^{3-}, P^{3-}
16	2−	O^{2-}, S^{2-}
17	1−	F^-, Cl^-, Br^-, I^-
Other fixed		Zn^{2+}, Ag^+

Worked examples: naming

Example: Name NaCl
Step 1: Name the metal. Na is sodium.
Step 2: Name the nonmetal with -ide. Cl is chlorine, which becomes chloride.
Step 3: No prefixes. Do not say "monosodium."
Answer: sodium chloride

Example: Name $CaBr_2$
Step 1: Name the metal. Ca is calcium.
Step 2: Name the nonmetal with -ide. Br is bromine, which becomes bromide.
Step 3: No prefixes. The subscript 2 is not stated in the name.
Answer: calcium bromide

Example: Name Li_3N
Step 1: Name the metal. Li is lithium.
Step 2: Name the nonmetal with -ide. N is nitrogen, which becomes nitride.
Step 3: No prefixes. The 3 is not stated.
Answer: lithium nitride

Worked examples: writing formulas

Example: Formula for magnesium oxide
Step 1: Write the ions with charges. Mg^{2+} and O^{2-}.
Step 2: Balance the charges. One Mg^{2+} gives 2+. One O^{2-} gives 2−. One of each already cancels.
Step 3: Write the subscripts. One of each, so no subscripts are written: MgO.
Step 4: Check charges. (2+) + (2−) = 0.
Answer: MgO

Example: Formula for aluminum sulfide
Step 1: Write the ions with charges. Al^{3+} and S^{2-}.
Step 2: Balance the charges. One Al^{3+} gives 3+, but one S^{2-} gives only 2−. Find the smallest charge both can reach: 6. Two Al give 2 × (3+) = 6+. Three S give 3 × (2−) = 6−.
Step 3: Write the subscripts. 2 Al and 3 S: Al_2S_3.
Step 4: Check charges. 6+ + 6− = 0.
Answer: Al_2S_3

Example: Formula for sodium sulfide
Step 1: Write the ions with charges. Na^+ and S^{2-}.
Step 2: Balance the charges. One S^{2-} gives 2−. Each Na^+ gives only 1+, so two Na are needed to reach 2+.
Step 3: Write the subscripts. 2 Na and 1 S (the 1 is not written): Na_2S.
Step 4: Check charges. 2(1+) + (2−) = 0.
Answer: Na_2S

2. Ionic Compounds with Polyatomic Ions

A polyatomic ion is a group of atoms that carries an overall charge and behaves as a single unit. The ion names and charges must be memorized.

Name	Formula	Name	Formula
Ammonium	NH_4^+	Hydroxide	OH^-
Nitrate	NO_3^-	Nitrite	NO_2^-
Sulfate	SO_4^{2-}	Sulfite	SO_3^{2-}
Phosphate	PO_4^{3-}	Phosphite	PO_3^{3-}
Carbonate	CO_3^{2-}	Hydrogen carbonate (bicarbonate)	HCO_3^-
Acetate	$C_2H_3O_2^-$	Cyanide	CN^-
Chlorate	ClO_3^-	Perchlorate	ClO_4^-
Chlorite	ClO_2^-	Hypochlorite	ClO^-
Permanganate	MnO_4^-	Chromate	CrO_4^{2-}
Dichromate	$Cr_2O_7^{2-}$	Peroxide	O_2^{2-}

Pattern for oxyanions (chlorine): per___ate (most O) → ___ate → ___ite → hypo___ite (least O). The charge stays the same; only the number of oxygens changes.

Naming

8. Name the cation, then the anion.
9. Keep the polyatomic ion name exactly as is. Do **not** change its ending to -ide.
10. If a monatomic ion is paired with a polyatomic ion, only the monatomic one takes -ide (for example, NH_4Cl is ammonium chloride).

Writing the formula

11. Write each ion with its charge.
12. Balance the charges: decide how many of each ion are needed so the total positive charge equals the total negative charge. Treat each polyatomic ion as one unit.
13. Write those numbers as subscripts. If a polyatomic ion is needed more than once, put it in **parentheses**: $(NO_3)_2$. Never change the subscripts inside the ion.
14. Check that the total positive and negative charges cancel.

Worked examples: naming

Example: Name K_2SO_4
Step 1: Name the cation, then the anion. K is potassium. SO_4 is a polyatomic ion.
Step 2: Keep the polyatomic name. SO_4^{2-} is sulfate (no -ide).
Answer: potassium sulfate

Example: Name $Ca(NO_3)_2$
Step 1: Name the cation. Ca is calcium.

Step 2: Find the polyatomic ion. The group in parentheses is NO_3^-, nitrate. The 2 outside the parentheses only tells how many, so it is not in the name.
Answer: calcium nitrate

Example: Name NH_4Cl
Step 1: Name the cation. NH_4^+ is ammonium (a polyatomic cation).
Step 2: Name the anion. Cl^- is a monatomic ion, so it takes -ide: chloride.
Answer: ammonium chloride

Worked examples: writing formulas

Example: Formula for ammonium phosphate
Step 1: Write the ions with charges. NH_4^+ and PO_4^{3-}.
Step 2: Balance the charges. One PO_4^{3-} gives 3−. Each NH_4^+ gives only 1+, so three NH_4 are needed to reach 3+.
Step 3: Write the subscripts. 3 NH_4 and 1 PO_4. Only NH_4 is repeated, so only it gets parentheses: $(NH_4)_3PO_4$.
Step 4: Check charges. 3(1+) + (3−) = 0.
Answer: $(NH_4)_3PO_4$

Example: Formula for aluminum hydroxide
Step 1: Write the ions with charges. Al^{3+} and OH^-.
Step 2: Balance the charges. One Al^{3+} gives 3+. Each OH^- gives only 1−, so three OH are needed to reach 3−.
Step 3: Write the subscripts. 1 Al and 3 OH. OH is repeated, so it goes in parentheses: $Al(OH)_3$.
Step 4: Check charges. (3+) + 3(1−) = 0.
Answer: $Al(OH)_3$

Example: Formula for calcium carbonate
Step 1: Write the ions with charges. Ca^{2+} and CO_3^{2-}.
Step 2: Balance the charges. One Ca^{2+} gives 2+. One CO_3^{2-} gives 2−. One of each already cancels.
Step 3: Write the subscripts. One of each, so no subscripts and no parentheses: $CaCO_3$.
Step 4: Check charges. (2+) + (2−) = 0.
Answer: $CaCO_3$

3. Ionic Compounds with Transition Metals

Most transition metals (and some others, such as Sn and Pb) can form more than one cation, so the name must show which charge the metal has. We use a Roman numeral in parentheses equal to the metal ion's charge.

Ion	Stock name	Ion	Stock name
Fe^{2+}	iron(II)	Fe^{3+}	iron(III)
Cu^{+}	copper(I)	Cu^{2+}	copper(II)
Sn^{2+}	tin(II)	Sn^{4+}	tin(IV)
Pb^{2+}	lead(II)	Pb^{4+}	lead(IV)
Co^{2+}	cobalt(II)	Co^{3+}	cobalt(III)
Mn^{2+}	manganese(II)	Cr^{3+}	chromium(III)
Hg^{2+}	mercury(II)	Hg_2^{2+}	mercury(I)

Exceptions with a single charge (no Roman numeral): Ag^{+}, Zn^{2+}, Cd^{2+}, and Al^{3+}.

Naming: find the metal's charge

15. Identify the anion and its charge (from the periodic table or polyatomic ion list).
16. Multiply the anion charge by its subscript to get the total negative charge.
17. The total positive charge must equal this. Divide by the number of metal atoms to get the charge on each metal ion.
18. Write the metal name with the Roman numeral, then the anion name.

Writing the formula

19. Write the metal ion using the charge given by the Roman numeral, and the anion with its charge.
20. Balance the charges: decide how many of each ion are needed so the total positive charge equals the total negative charge.
21. Write those numbers as subscripts (use parentheses for a repeated polyatomic ion).
22. Check that the total positive and negative charges cancel.

Worked examples: naming

Example: Name $FeCl_3$
Step 1: Anion and charge. Cl^{-} (chloride), 1−.
Step 2: Total negative charge. 3 × (1−) = 3−.
Step 3: Charge on each metal. One Fe must balance 3−, so Fe is 3+.
Step 4: Write the name. Iron(III) plus chloride.
Answer: iron(III) chloride

Example: Name Cu_2O
Step 1: Anion and charge. O^{2-} (oxide), 2−.
Step 2: Total negative charge. 1 × (2−) = 2−.
Step 3: Charge on each metal. Two Cu must total 2+, so each is 2+ ÷ 2 = 1+.
Step 4: Write the name. Copper(I) plus oxide.
Answer: copper(I) oxide

Example: Name $Fe_2(SO_4)_3$

Step 1: Anion and charge. SO_4^{2-} (sulfate), 2−.
Step 2: Total negative charge. 3 × (2−) = 6−.
Step 3: Charge on each metal. Two Fe must total 6+, so each is 6+ ÷ 2 = 3+.
Step 4: Write the name. Iron(III) plus sulfate.
Answer: iron(III) sulfate

Worked examples: writing formulas

Example: Formula for lead(IV) oxide
Step 1: Write the ions with charges. Roman numeral IV means Pb^{4+}. Oxide is O^{2-}.
Step 2: Balance the charges. One Pb^{4+} gives 4+. Each O^{2-} gives 2−, so two O are needed to reach 4−.
Step 3: Write the subscripts. 1 Pb and 2 O: PbO_2.
Step 4: Check charges. (4+) + 2(2−) = 0.
Answer: PbO_2

Example: Formula for copper(II) nitrate
Step 1: Write the ions with charges. Cu^{2+} and NO_3^-.
Step 2: Balance the charges. One Cu^{2+} gives 2+. Each NO_3^- gives 1−, so two NO_3 are needed to reach 2−.
Step 3: Write the subscripts. 1 Cu and 2 NO_3. NO_3 is repeated, so it goes in parentheses: $Cu(NO_3)_2$.
Step 4: Check charges. (2+) + 2(1−) = 0.
Answer: $Cu(NO_3)_2$

Example: Formula for iron(III) oxide
Step 1: Write the ions with charges. Fe^{3+} and O^{2-}.
Step 2: Balance the charges. One Fe^{3+} gives 3+, but one O^{2-} gives only 2−. Find the smallest charge both can reach: 6. Two Fe give 2 × (3+) = 6+. Three O give 3 × (2−) = 6−.
Step 3: Write the subscripts. 2 Fe and 3 O: Fe_2O_3.
Step 4: Check charges. 2(3+) + 3(2−) = 0.
Answer: Fe_2O_3

4. Covalent (Molecular) Compounds

Covalent compounds contain two nonmetals. Because nonmetals can combine in several ratios, **prefixes** show how many atoms of each element are present.

Number	Prefix	Number	Prefix
1	mono-	6	hexa-
2	di-	7	hepta-
3	tri-	8	octa-
4	tetra-	9	nona-

5	penta-	10	deca-

Naming

23. Name the first element with a prefix for its count, **except** omit mono- on the first element.
24. Name the second element with a prefix (always, including mono-) and the ending **-ide**.
25. Drop the final a or o of the prefix when the element name begins with a vowel (monoxide, pentoxide, not monooxide, pentaoxide).

Writing the formula

26. Write the symbols in the same order as in the name.
27. Turn each prefix into a subscript (no prefix on the first element means 1).
28. **Do not** balance charges or reduce.

Worked examples: naming

Example: Name CO_2
Step 1: First element. One C, so no prefix: carbon.
Step 2: Second element. Two O, so di- plus oxide: dioxide.
Answer: carbon dioxide

Example: Name CO
Step 1: First element. One C, no prefix: carbon.
Step 2: Second element. One O, so mono- is used on the second element, and the a drops before a vowel: monoxide.
Answer: carbon monoxide

Example: Name N_2O_4
Step 1: First element. Two N: dinitrogen (the first element uses a prefix when the count is above 1).
Step 2: Second element. Four O: tetra- plus oxide, and the a drops: tetroxide.
Answer: dinitrogen tetroxide

Example: Name PCl_5
Step 1: First element. One P, no prefix: phosphorus.
Step 2: Second element. Five Cl: penta- plus chloride.
Answer: phosphorus pentachloride

Worked examples: writing formulas

Example: Formula for sulfur hexafluoride
Step 1: Write the symbols in order. S then F.
Step 2: Prefixes to subscripts. Sulfur has no prefix, so 1. Hexa- means 6.

Answer: SF_6

Example: Formula for diphosphorus pentoxide
Step 1: Write the symbols in order. P then O.
Step 2: Prefixes to subscripts. Di- means 2, and penta- means 5.
Answer: P_2O_5 (do not reduce)

Example: Formula for carbon tetrachloride
Step 1: Write the symbols in order. C then Cl.
Step 2: Prefixes to subscripts. No prefix means 1, and tetra- means 4.
Answer: CCl_4

Note: some covalent compounds have common names used instead: H_2O (water), NH_3 (ammonia), CH_4 (methane).

5. Acids

An acid is a compound that releases H^+ in water. Acid formulas are written with H first, followed by an anion. **The acid name comes from the name of that anion.**

The one rule to remember

Look at the ending of the anion, then swap it as shown:

If the anion ends in...	The acid name is...	Example (anion → acid)
-ide	hydro- + root + -ic acid	chloride → hydrochloric acid (HCl)
-ate	root + -ic acid	nitrate → nitric acid (HNO_3)
-ite	root + -ous acid	nitrite → nitrous acid (HNO_2)

Memory aid: "I ate something icky." -ate becomes -ic, so -ite becomes -ous. Only the -ide acids get the hydro- prefix, and they contain no oxygen.

Root note: for sulfur and phosphorus the root is longer: sulfate → sulfuric, phosphate → phosphoric.

Naming an acid from its formula

29. Find the anion. It is everything after the H. (HCl: Cl^-. H_2SO_4: SO_4^{2-}.)
30. Name the anion, including its ending (chloride, sulfate).
31. Look up the ending in the table above and build the acid name.
32. Write "acid" as a separate word.

Writing an acid formula from its name

33. Work backward using the table: hydro-…-ic means the anion ends in -ide; -ic means -ate; -ous means -ite.
34. Write the anion and its charge.
35. Write H first. The number of H^+ needed equals the size of the anion charge, so that the total charge is zero.
36. Do not write the charges in the final formula.

Worked examples: naming

Example: Name HBr
Step 1: Find the anion. After H is Br^-.
Step 2: Name the anion. Bromide, which ends in -ide.
Step 3: Use the table. -ide → hydro- + root + -ic: hydrobromic.
Answer: hydrobromic acid

Example: Name H_2SO_4
Step 1: Find the anion. After H_2 is SO_4^{2-}.
Step 2: Name the anion. Sulfate, which ends in -ate.
Step 3: Use the table. -ate → root + -ic (keep the longer root for sulfur): sulfuric.
Answer: sulfuric acid

Example: Name $HClO_2$
Step 1: Find the anion. After H is ClO_2^-.
Step 2: Name the anion. Chlorite, which ends in -ite.
Step 3: Use the table. -ite → root + -ous: chlorous.
Answer: chlorous acid

Worked examples: writing formulas

Example: Formula for phosphoric acid
Step 1: Work backward. -ic means the anion ends in -ate: phosphate.
Step 2: Write the anion and charge. PO_4^{3-}.
Step 3: Add H. The anion has a 3− charge, so three H^+ are needed.
Answer: H_3PO_4

Example: Formula for hydrosulfuric acid
Step 1: Work backward. hydro-…-ic means the anion ends in -ide: sulfide.
Step 2: Write the anion and charge. S^{2-}.
Step 3: Add H. The anion has a 2− charge, so two H^+ are needed.
Answer: H_2S

Example: Formula for nitrous acid

Step 1: Work backward. -ous means the anion ends in -ite: nitrite.
Step 2: Write the anion and charge. NO_2^-.
Step 3: Add H. The anion has a 1− charge, so one H^+ is needed.
Answer: HNO_2

Quick Decision Guide

Compound type	Clue	Naming rule
Binary ionic	Metal + nonmetal	metal + nonmetal-ide
With polyatomic ions	Contains a polyatomic ion	use ion names as they are
With transition metals	Metal that has multiple charges	Roman numeral for the metal charge
Covalent	Two nonmetals	prefixes, no reduction
Acid	H first, in water	hydro-ic, -ic, or -ous